

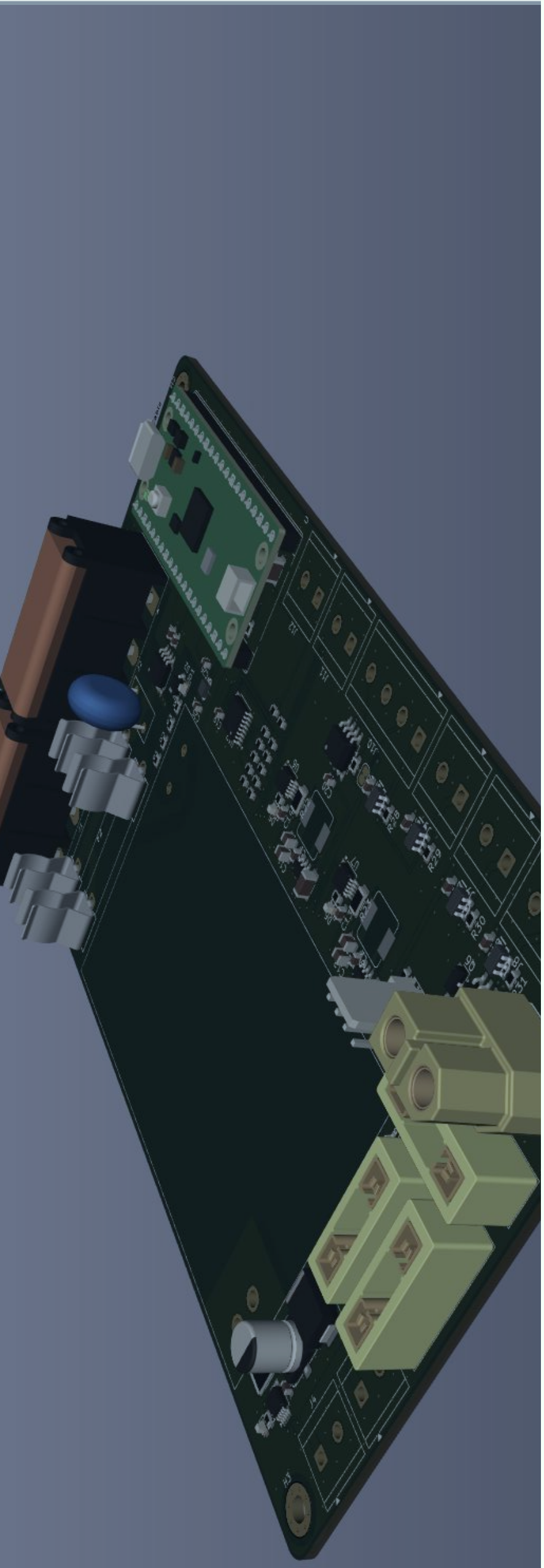
STUDENT-BUILT SEMICONDUCTOR FAB · VANCOUVER, BC

Vancouver Hacker Fab Sponsorship Package 2026–2027

We design, build and open-source the tools to fabricate semiconductor devices in Vancouver. This year, we are making the first student-fabricated transistor in Vancouver.

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Our Team

Vancouver Hacker Fab is a team run by students at the University of British Columbia building an open-source semiconductor fab. We design, build, document and share DIY machines and processes for making chips, so that undergraduates can learn how microchips are made while gaining hands on experience.

We started in October 2024 as first-year engineering students working out of our dorm rooms. Since then we have built a maskless lithography stepper, raised US$5,000 in grants, demoed our work at UBC Imagine Day, and grown into a multidisciplinary team with sub-teams for lithography, deposition, annealing, etching and testing.

We are part of the global Hacker Fab network, which began at Carnegie Mellon University in 2022 and now includes teams at Waterloo, UC Irvine, Ohio State and MIT.

2024
Founded at UBC

US$5K
Raised in grants

5
Fab tools in development

~10 µm
Target feature size



Students on our team gain hands-on experience in:

- Optics and precision motion control
- Vacuum, RF and plasma systems
- PCB design and embedded firmware
- CAD, machining and 3D printing
- Thin-film and semiconductor process development
- Technical documentation and project leadership

Founded by Anush Mutyala (Schulich Leader ’24, Computer Engineering) and Marwa Attaii (DAW International Scholar ’24, Materials Engineering).

Our Tools

Making a transistor takes four core process steps: patterning, deposition, annealing and etching. Each one needs its own machine, and we design and build every one ourselves from accessible, off-the-shelf parts.

STEP 1 · PATTERNING

BUILT · UPGRADING

Maskless Lithography Stepper

Projects circuit patterns onto photoresist-coated wafers using a DLP projector and precision stages. It targets features near 10 µm without the cost of physical photomasks.

STEP 1 · PATTERNING

OPERATIONAL

Spin Coater

Spins wafers at controlled speeds to coat them with thin, uniform films of photoresist before exposure.

STEP 2 · ANNEALING & DOPING

OPERATIONAL

Tube Furnace

A programmable high-temperature furnace for growing oxides, diffusing dopants and annealing. Our first milestone is a working PN diode.

STEP 3 · DEPOSITION

IN DEVELOPMENT

RF Sputtering Chamber

A vacuum chamber that uses RF plasma to deposit thin films of metals and oxides, the conductive and insulating layers of a transistor.

STEP 4 · ETCHING

PLANNED

Plasma Etcher

Uses ionized gas to etch fine features into the wafer, completing the toolset we need to make transistors in-house.

Road to the First Transistor

OCT 2024

Team founded

Two first-year students set out to make the first student-fabricated chip in Vancouver.

2025

First grants, first tool

US$5,000 from Emergent Ventures and the 1517 Fund paid for our maskless lithography stepper.

SEP 2025

Imagine Day and team growth

We demoed our tools publicly and recruited new members into tool-focused sub-teams.

2026

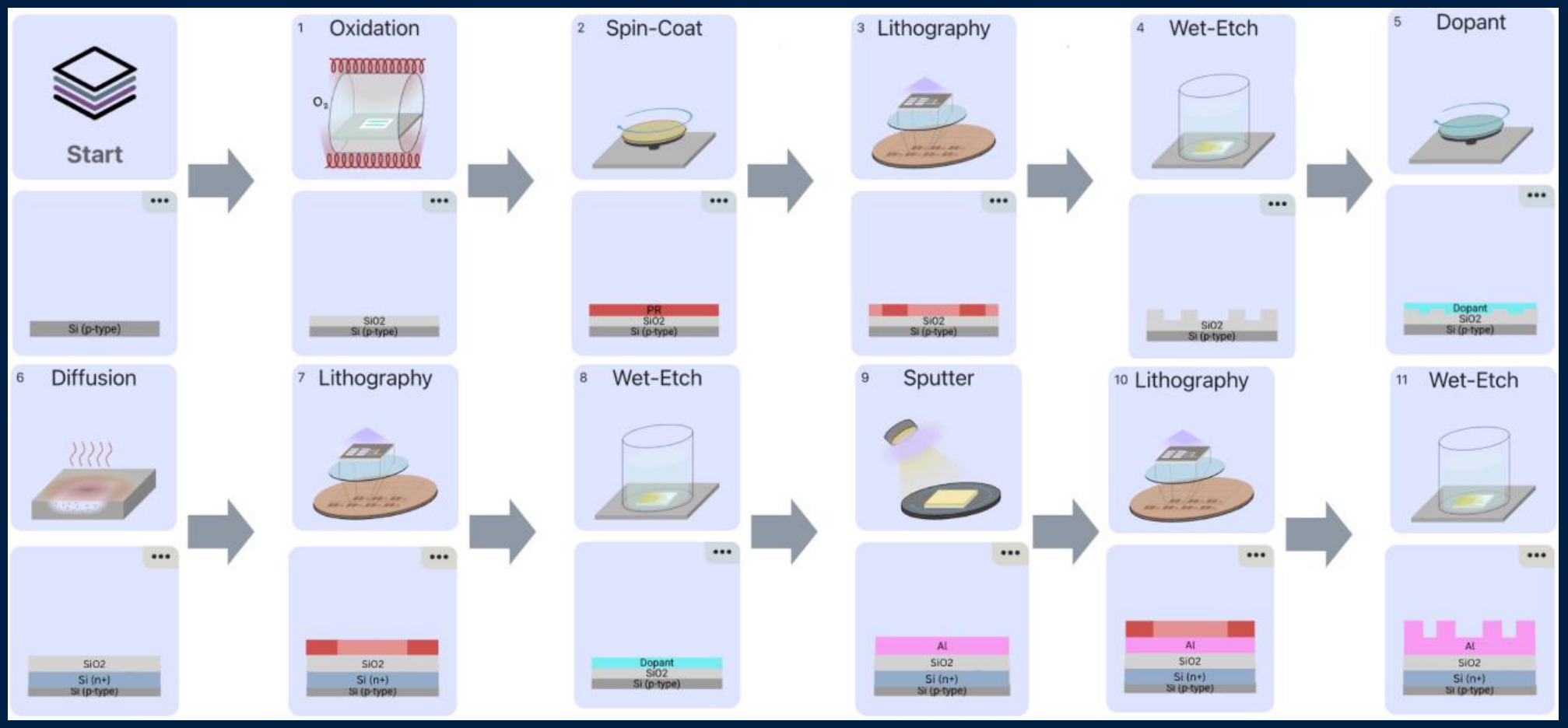
Building out the fab

Work began on the tube furnace and the sputtering vacuum system, alongside process development.

2026–2027 GOAL

First transistor at UBC

A PN diode, then an NMOS transistor characterized with I–V measurements, made entirely on tools we built.



Simplified process flow for fabrication of a self-aligned NMOS transistor

Why It Matters

The gap

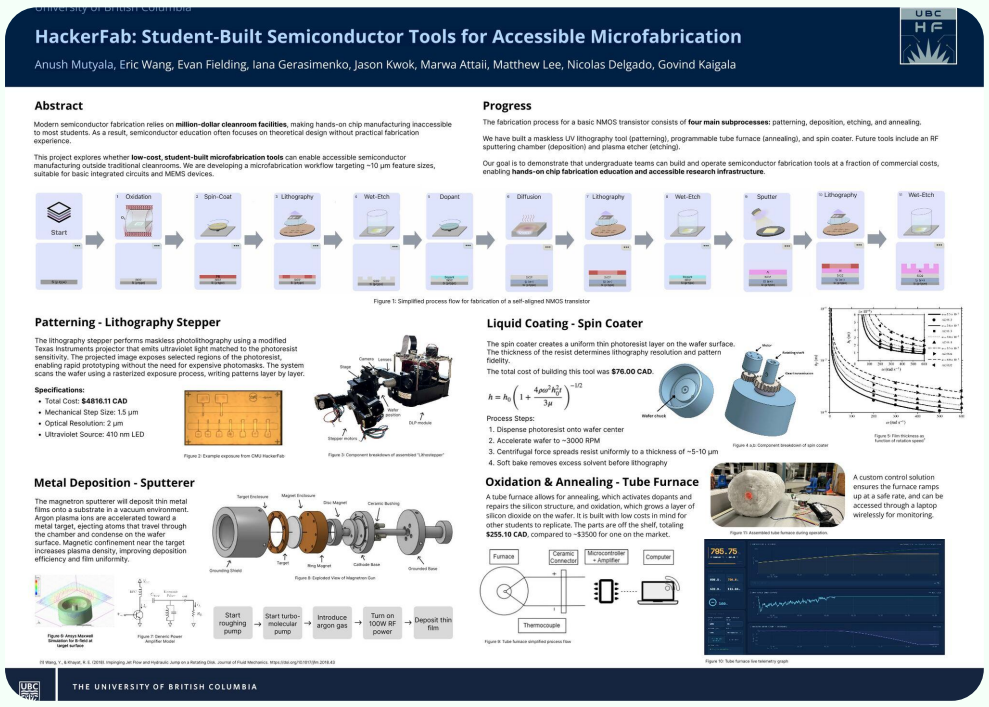
UBC has world-class nanofabrication facilities, but access is mostly reserved for graduate research and senior courses. Undergraduates who want to learn how chips are made have almost no hands-on path.

Our answer

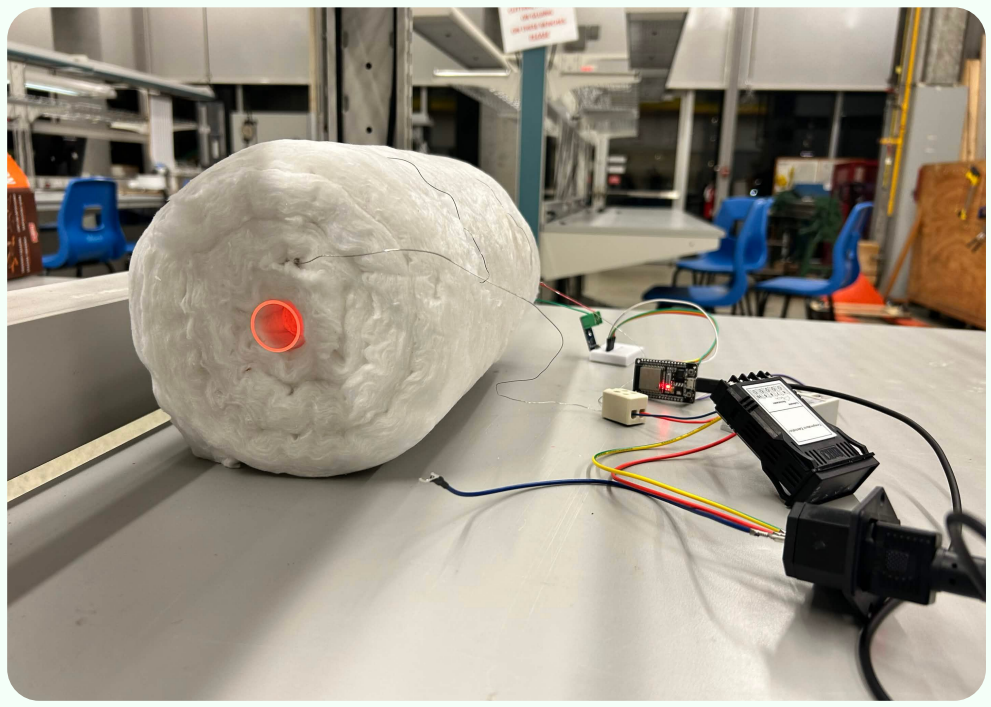
Low-cost tools built from off-the-shelf parts, fully documented and open source, so any student, or any other university, can learn from and replicate them.

In the community

We share what we build. We demo our tools at UBC Imagine Day, run info sessions that introduce students to how chips are made, and publish our designs back to the global Hacker Fab network. Additionally, we presented our work at MURC 2026 at UBC.



MURC · UBC



Tube furnace prototype

Canada needs a semiconductor workforce. Our members graduate having built fab tools with their own hands.

Where Your Money Goes

Our 2026–2027 plan centres on finishing the toolset needed to make a transistor. As a sponsor, you directly fund hardware, materials and hands-on training for the students who build it.

Plasma etcher	ETCHING	$15,000
RF sputterer	DEPOSITION	$2,000 (est.)
Lithography stepper upgrades	PATTERNING	$1,000 (est.)
Wafers, photoresist & chemicals	MATERIALS	$1,000 (est.)
Tube furnace	ANNEALING & DOPING	$300 (est.)
Spin coater	COATING	$200 (est.)

TOTAL 2026–2027 NEED

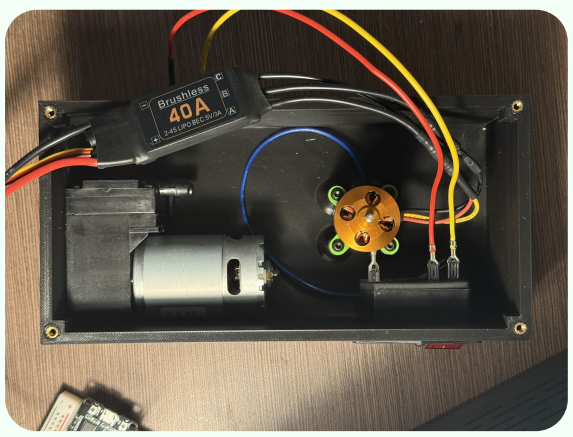
Amounts in CAD. Estimates are based on our project roadmap.

$19,500

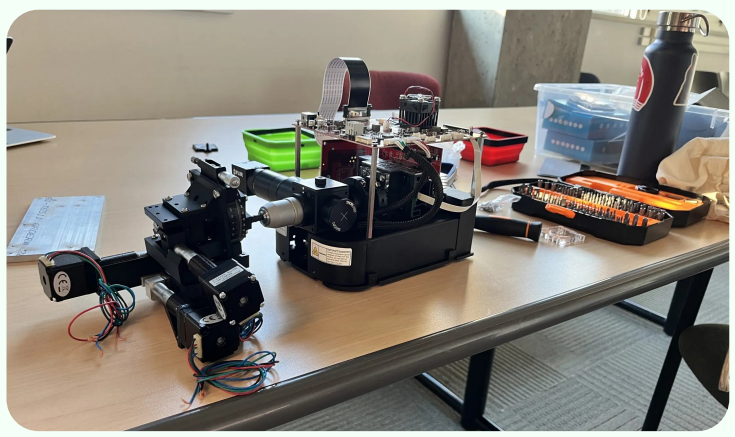
Sponsorship Levels

We welcome both monetary and in-kind support. If you would like to partner with us in a different way, we are happy to build a package around what matters to you.

Benefit	Platinum $2,500+	Gold $1,000–2,499	Silver $500–999	Bronze Up to $500
Logo on website and tool builds	**Largest**	Large	Medium	Small
Logo on posters, merch and conference materials	**Largest**	Large	Medium	Small
Dedicated social media feature	✓	✓	✓	—
Live tool demo	✓	✓	—	—



SPIN COATER V1



LITHOSTEPPER V1

In-kind sponsorship

In-kind sponsors are recognized at the tier matching the value of their contribution. We especially need vacuum components, RF power supplies, silicon wafers, photoresist, PCB fabrication and lab instruments.

Thank You to Our Supporters

Our first tools were made possible by early grants from people who believed in a student-built chip fab.

MERCATUS CENTER · $4K USD GRANT

EMERGENT VENTURES

1517 · $1K USD GRANT

1517
Medici Project

2026–2027 sponsors

YOUR LOGO HERE

YOUR LOGO HERE

YOUR LOGO HERE

YOUR LOGO HERE

YOUR LOGO HERE

YOUR LOGO HERE

With thanks to Hacker Fab at Carnegie Mellon University for open-sourcing the tools and processes we build on. We are also thankful for support from faculty and staff in the School of Biomedical Engineering at UBC.

hackerfab

UBC SBME



10 — GET IN TOUCH

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CONTACT	Anush Mutyala, Co-founder

Help us make the first student-fabricated transistor in Vancouver.